Relevant passage ranking

Alpha Follows the Hops

Query Complexity and Score Fusion in Graph-Augmented Re-ranking

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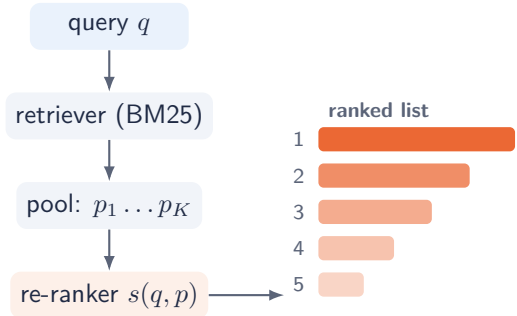


Part 1

Background

The task: rank passages for a query

- Input: a query q and a pool of candidate passages $p_1, \ldots, p_K$.
- Output: the passages ordered by relevance to q.
- A **scoring function** gives each pair (q, p_i) a score $s(q, p_i)$; the ranking sorts by it.
- Usual pipeline: a cheap retriever fills the pool (BM25, top-K), an expensive model *re-ranks* it.
- Evaluated with nDCG@10 and MRR@10.



Scoring functions: one score per pair

TF-IDF

$\sum_{t \in q} \mathrm{tf}(t, p) \cdot \mathrm{idf}(t)$

Exact term overlap, weighted by rarity.

BM25

$\sum_{t \in q} \mathrm{idf}(t) \, \dfrac{\mathrm{tf}\,(k_1 + 1)}{\mathrm{tf} + k_1 B_p}$

TF saturation and length normalisation; the standard retriever.

Bi-encoder

$\cos\big(E(q),\, E(p)\big)$

Query and passage encoded separately; fast, indexable.

Cross-encoder

$\mathrm{BERT}\big([q\,;\,p]\big)$

Joint encoding; most accurate, too slow to index.

lexical $\longrightarrow$ neural: more accurate, more expensive

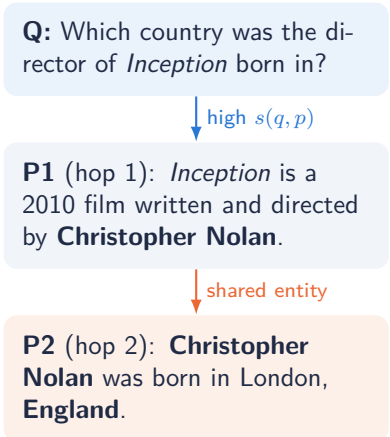
All of them score each passage **on its own**: $s(q, p_i)$ never looks at the other candidates.

Where it breaks: multi-hop questions

Single-hop: one passage answers the query.

Multi-hop: the answer needs a **chain** of passages.

- The second passage (about the director) shares almost no words with the query.
- A pointwise scorer ranks it low, although it is needed.
- Its relevance shows only *through its link* to P1.



Illustrative example.

GAR: Graph-based Adaptive Re-ranking

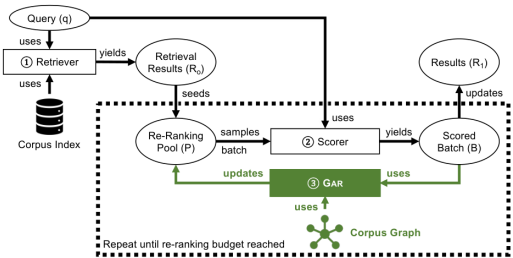


Figure 1: Overview of GAR. Traditional re-ranking exclusively scores results seeded by the retriever. GAR (in green) adapts the re-ranking pool after each batch based on the computed scores and a pre-computed graph of the corpus.

Screenshot from MacAvaney et al. (CIKM 2022).

- **Clustering hypothesis**: passages close to a relevant passage tend to be relevant too.
- Offline: a **corpus graph**, each passage linked to its k most similar passages.
- Online: after each scored batch, the *neighbours* of the top passages join the pool.
- Fixes the recall limit of the first-stage pool; the scorer itself stays pointwise.

GAR: the re-ranking loop

Algorithm 1 Graph-based Adaptive Re-Ranking

Input: Initial ranking R_0, batch size b, budget c, corpus graph G
Output: Re-Ranked pool R_1

$R_1 \leftarrow \emptyset$ ▷ Re-Ranking results
$P \leftarrow R_0$ ▷ Re-ranking pool
$F \leftarrow \emptyset$ ▷ Graph frontier
do
  $B \leftarrow \textsc{Score}(\text{top } b \text{ from } P, \text{subject to } c)$ ▷ e.g., monoT5
  $R_1 \leftarrow R_1 \cup B$ ▷ Add batch to results
  $R_0 \leftarrow R_0 \setminus B$ ▷ Discard batch from initial ranking
  $F \leftarrow F \setminus B$ ▷ Discard batch from frontier
  $F \leftarrow F \cup (\textsc{Neighbours}(B, G) \setminus R_1)$ ▷ Update frontier
  $P \leftarrow \begin{cases} R_0 & \text{if } P = F \\ F & \text{if } P = R_0 \end{cases}$ ▷ Alternate initial ranking and frontier
while $|R_1| < c$
$R_1 \leftarrow R_1 \cup \textsc{Backfill}(R_0, R_1)$ ▷ Backfill remaining items

Algorithm 1 of MacAvaney et al. (CIKM 2022).

- Alternates between the initial ranking R_0 and the *graph frontier* F.
- Up to $+8\%$ nDCG for BM25 $\gg$ monoT5 on MS MARCO.
- The graph decides **which** passages get scored, not **how** they are scored.

Next step: let the graph change the *score* itself, so a passage is judged together with its neighbours $\Rightarrow$ ReGAT-BERT.

ReGAT-BERT: the idea

Semantic score s_{sem}

A BERT cross-encoder reads (q, p) jointly.
Strong on *direct* matches.

Contextual score s_{ctx}

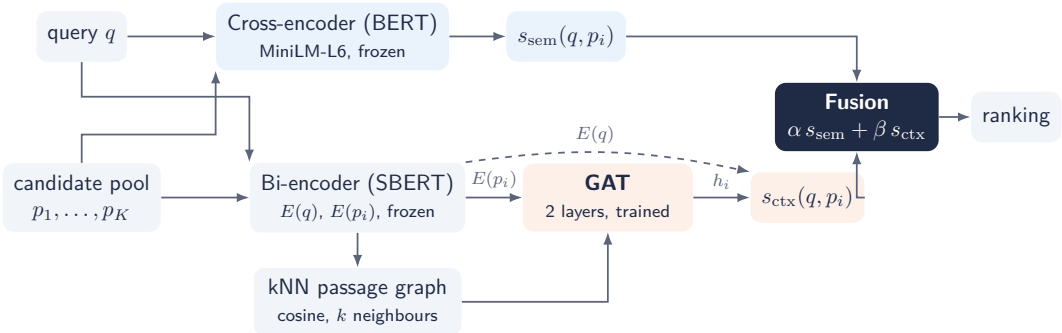
Passages form a graph; a Graph Attention Network refines each passage embedding with its neighbours.
Scores the passage *in context*.

- Two complementary views of relevance.
- Final score = a weighted sum of the two:

$$\text{score}(q, p) = \alpha\, s_{\text{sem}} + \beta\, s_{\text{ctx}}$$

- α, β are learned **once**, then the **same weights apply to every query**.

ReGAT-BERT architecture

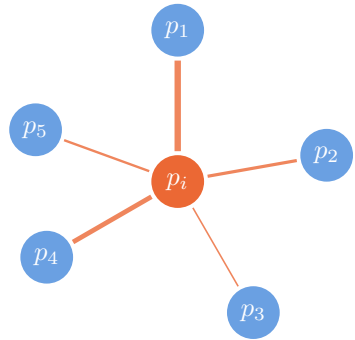


Our diagram of the pipeline as reproduced in this project (models as used in our experiments).

GAT step 1: build a graph over the pool

One graph per query, over its K candidate passages.

node i	candidate passage p_i ($K = 100$ on MS MARCO, 10–20 in closed pools)
$x_i = E(p_i) \in \mathbb{R}^d$	node feature: the frozen bi-encoder embedding of p_i, $d = 384$
edge $j \to i$	p_j is one of the k nearest neighbours of p_i by cosine, and $\cos(x_i, x_j) \geq 0.3$; $k = 8$ (MS MARCO), 3 (multi-hop)
self-loop	edge $i \to i$, so a passage keeps its own information
$\mathcal{N}(i)$	neighbourhood of i: its neighbours and itself
X, A	all node features $[x_1, \ldots, x_K]$ and the adjacency matrix ($A_{ij} = 1$ if edge)



edge width $\propto$ attention π_{ij} (step 2): the GAT learns which neighbours matter

GAT step 2: how much should p_i listen to p_j?

Attention score (GATv2, one head):

$$e_{ij} = \mathbf{a}^\top \,\mathrm{LeakyReLU}\big(\mathbf{W}_t\, x_i + \mathbf{W}_s\, x_j\big)$$

Attention weight: softmax over the neighbourhood

$$\pi_{ij} = \frac{\exp(e_{ij})}{\sum_{k \in \mathcal{N}(i)} \exp(e_{ik})}$$

$\pi_{ij} \geq 0$, $\sum_j \pi_{ij} = 1$.

Called α_{ij} in the GAT papers; renamed to avoid a clash with the fusion weight α.

x_i, x_j	embedding of the passage being updated (i) and of a neighbour (j)
$\mathbf{W}_t, \mathbf{W}_s$	learned $d' \times d$ projections for the target (i) and source (j) node
LeakyReLU	$\max(z, 0.2z)$ per coordinate (non-linearity)
$\mathbf{a} \in \mathbb{R}^{d'}$	learned attention vector; $\mathbf{a}^\top(\cdot)$ is a dot product: vector $\rightarrow$ *one number*
e_{ij}	raw importance of neighbour j for i
π_{ij}	normalised attention weight

GATv2 applies $\mathbf{a}$ *after* the non-linearity ("dynamic attention").

GAT step 3: aggregate, multi-head, two layers

One head: weighted average of projected neighbours

$$x'_i = \sum_{j \in \mathcal{N}(i)} \pi_{ij} \, \mathbf{W}_s \, x_j + \mathbf{b}$$

Layer 1: $M = 4$ heads, outputs concatenated

$$h_i^{(1)} = \Big\|_{m=1}^{4} \Big(\sum_j \pi_{ij}^{(m)} \mathbf{W}_s^{(m)} x_j + \mathbf{b}^{(m)} \Big) \in \mathbb{R}^{4 \cdot 128}$$

Layer 2: one head, back to dimension d

$$h_i^{(2)} = \sum_j \pi_{ij}^{(2)} \, \mathbf{W}_s^{(2)} \, \mathrm{ELU}\big(h_j^{(1)}\big) + \mathbf{b}^{(2)} \in \mathbb{R}^{384}$$

$\mathbf{b}$	learned bias vector
m	head index; each head has its own $\mathbf{W}$, $\mathbf{a}$, so it can attend to a different kind of relation
$\|$	concatenation of the 4 head outputs ($4 \times 128 = 512$ numbers)
$\mathrm{ELU}(z)$	z if $z > 0$, else $e^z - 1$ (smooth non-linearity between layers)
dropout	0.1 on the attention weights while training

2 layers = 2 graph hops: p_i also hears its neighbours' neighbours.

GAT step 4: refined passage → contextual score

Residual: keep the meaning, add a correction

$$z_i = x_i + \mathrm{GAT}(X, A)_i$$

LayerNorm, then unit length

$$\mathrm{LN}(z) = \boldsymbol{\gamma} \odot \frac{z - \mu_z}{\sqrt{\sigma_z^2 + \epsilon}} + \boldsymbol{\eta}$$

$$h_i = \mathrm{LN}(z_i) \,/\, \|\mathrm{LN}(z_i)\|_2$$

Contextual score

$$s_{\mathrm{ctx}}(q, p_i) = E(q)^\top h_i = \cos\big(E(q), h_i\big)$$

$\mathrm{GAT}(X, A)_i$	2-layer GAT output $h_i^{(2)}$ for node i, given all features X and the graph A
LN	Layer Normalisation: re-centres and re-scales one vector over its own d coordinates
μ_z, σ_z	mean and std of the d coordinates of z
$\boldsymbol{\gamma}$, $\boldsymbol{\eta}$	learned scale and shift ($\in \mathbb{R}^d$); $\odot$ = element-wise product
$\|\cdot\|_2$	Euclidean length: h_i becomes a unit vector
$E(q)$	unit-length query embedding from the *same* bi-encoder

h_i and $E(q)$ share one $\mathbb{R}^{384}$ space: the residual starts h_i at the bi-encoder vector x_i.

Training the GAT: InfoNCE loss

$$\mathcal{L}_{\mathrm{GAT}} = -\log \frac{\sum_{p^+} \exp\left(s_{\mathrm{ctx}}(q, p^+)/\tau\right)}{\sum_{j=1}^{K} \exp\left(s_{\mathrm{ctx}}(q, p_j)/\tau\right)}$$

- The fraction is the share of the softmax over the pool that falls on the gold passages.
- Minimising $\mathcal{L}$ pulls gold h_i towards $E(q)$ and pushes the other pool passages away.
- Trained: $\mathbf{W}_s, \mathbf{W}_t, \mathbf{a}, \mathbf{b}, \boldsymbol{\gamma}, \boldsymbol{\eta}$. Bi-encoder and cross-encoder stay **frozen**.
- AdamW, lr 10^{-4}, early stopping on val nDCG@10; trained **once** per dataset, then frozen for all fusion experiments.

InfoNCE	a contrastive loss: cross-entropy of picking the gold passages out of the pool
p^+	gold (relevant) passages of q in the pool
p_j, K	all pool passages, pool size
$\tau = 0.05$	temperature: dividing by a small τ sharpens the softmax, so small score gaps matter
exp, log	natural exponential and logarithm

ReGAT-BERT fusion strategy

$$\text{score}(q,p) = \alpha \cdot s_{\text{sem}}(q,p) + \beta \cdot s_{\text{ctx}}(q,p)$$

α, β: two free scalars, learned with a pointwise **BCE** loss:

$$\mathcal{L}_{\text{BCE}} = -\frac{1}{N} \sum_{(q,p)} \Big[y \log \sigma(\text{score}) + (1-y) \log \big(1 - \sigma(\text{score})\big) \Big]$$

BCE	Binary Cross-Entropy: penalises a confident wrong relevance probability
$y \in \{0,1\}$	relevance label of the pair (q,p)
$\sigma(z)$	sigmoid $1/(1+e^{-z})$: score $\to$ probability
N	number of training pairs, each judged on its own

- Raw scores are fused as they are: the cross-encoder logit and the cosine live on **different scales**.
- One pair (α, β) for **all queries**: a keyword query and a 4-hop question get the same mix.

Part 2

Our study

Problem statement

Should the weight between the semantic and the contextual score be **the same for every query**, or should it depend on the query and its passages?

Simple / single-hop queries

One passage answers it; the cross-encoder sees everything it needs.

Expect $\alpha > 0.5$ (trust s_{sem}).

Complex / multi-hop queries

Evidence is spread over linked passages; only the graph sees the links.

Expect $\alpha < 0.5$ (trust s_{ctx}).

Hypothesis: α should *fall as the number of hops rises*, so a query-dependent $\alpha(\cdot)$ should beat a single fixed α.

Our score and the normalisers

$$\mathrm{score}(q, p_i) = \alpha \cdot \tilde{s}_{\mathrm{sem}}(q, p_i) + (1 - \alpha) \cdot \tilde{s}_{\mathrm{ctx}}(q, p_i), \qquad \alpha \in (0, 1)$$

$\tilde{s}$ = score normalised *within each query's pool*, so α is a true mixing weight. $\alpha = 1$: cross-encoder only; $\alpha = 0$: GAT only.

None

$\tilde{s}_i = s_i$

Raw scores (paper setting).

Min–max

$\tilde{s}_i = \frac{s_i - \min s}{\max s - \min s}$

Maps to $[0, 1]$; our primary normaliser.

Z-score

$\tilde{s}_i = \frac{s_i - \mu_q}{\sigma_q}$

Zero mean, unit variance per query.

Rank

$\tilde{s}_i = 1 - \frac{\mathrm{rank}_i - 1}{K - 1}$

Only the order counts; scale-free.

s_i: raw score of the i-th passage in the query's pool. $\min s$, $\max s$, μ_q, σ_q: minimum, maximum, mean and standard deviation over that pool. rank_i: position of p_i when the pool is sorted by s ($1 =$ best). K: pool size.

Modelling α: the gate network

$$\alpha(x) = \epsilon + (1 - 2\epsilon) \cdot \sigma\Big(\mathbf{w}_2^\top \operatorname{ReLU}\big(\mathbf{W}_1 x + \mathbf{b}_1\big) + b_2\Big)$$

Gate	input x (what α may depend on)
fixed	none: one learned scalar for all queries
Q	query embedding $E(q) \in \mathbb{R}^{384}$
S	12 score statistics $f(s_{\text{sem}}, s_{\text{ctx}})$
QS	$[E(q) \,\|\, f]$
QS+C	$[E(q) \,\|\, f \,\|\, c]$, c = length, #entities

Every query gets its **own** α, computed from x at test time; no labels are used.

x	gate input for this query (left)
$\mathbf{W}_1, \mathbf{b}_1$	hidden layer, 64 units (dropout 0.1)
$\operatorname{ReLU}(z)$	$\max(z, 0)$
$\mathbf{w}_2, b_2$	output layer: one number (a logit)
$\sigma(z)$	sigmoid, maps the logit into $(0, 1)$
$\epsilon = 10^{-6}$	keeps α strictly inside $(0, 1)$
$\|$	concatenation of input vectors

Gate inputs and training

S features (12 per query, from raw scores):

For each of s_{sem} and s_{ctx} over the pool:

mean, std	level and spread of the scores
top-1	best score
top-1 − top-2	margin: how confident the scorer is
entropy	of softmax(scores): how peaked

plus Spearman $\rho(s_{\text{sem}}, s_{\text{ctx}})$ (do the two rankings agree?) and the overlap of their top-10.

C: query length in words and number of named entities (spaCy).

Listwise loss with a pull towards α_0:

$$\mathcal{L} = -\log \sum_{p^+} \text{softmax}_i\left(\frac{\text{score}_i}{T}\right) + \lambda\,(\alpha - \alpha_0)^2$$

score_i	fused score of pool passage i
softmax_i	over the query's pool; first term = probability mass on gold p^+
T	learned temperature
α_0	best single fixed α on the validation grid (step 0.05)
$\lambda = 0.01$	strength of the pull towards α_0

$b_2 = \text{logit}(\alpha_0)$, $\mathbf{w}_2 \approx 0$ at start: every gate starts as the fixed baseline. AdamW, early stopping on val, 5 seeds. CE and GAT frozen.

Datasets: from single-hop to multi-hop

MS MARCO
single-hop
Real Bing queries; 1 relevant passage per query.

HotpotQA
2-hop
Bridge and comparison questions.

BEIR-HQA
2-hop, open
HotpotQA queries over 5M Wikipedia passages.

2Wiki
2–4 hops
4 types: comparison, compositional, inference, bridge-comparison.

MuSiQue
2, 3, 4 hops
Built by composing single-hop questions; hops labelled.

Dataset	Setting	K	GAT train	Fusion train	Val	Test	Gold/pool	Complexity labels
MS MARCO (A)	open (BM25)	100	30,000	10,000	2,000	6,980	–	proxies only
MS MARCO (B)	open (BM25)	100	1,870	1,150	698	1,745	–	proxies only
MuSiQue	closed pool	20	10,000	7,000	2,938	2,417	2.65	hops 2/3/4
HotpotQA	closed pool	10	20,000	10,000	2,000	7,405	2.00	bridge/comparison
BEIR-HotpotQA	open (BM25)	100	20,000	10,000	2,000	7,405	1.59	bridge/comparison
2Wiki	closed pool	10	20,000	10,000	2,000	12,576	2.44	4 types

Table 1: Datasets and splits (queries). Val is always carved from the training data; test is the official dev set (MS MARCO A: dev/small; MuSiQue, HotpotQA, 2Wiki: dev; BEIR: test). MS MARCO B splits dev/small itself, so the cross-encoder never saw any of its queries. Gold/pool = mean number of relevant passages among the K test candidates.

Part 3

Results

1. The best α differs by dataset

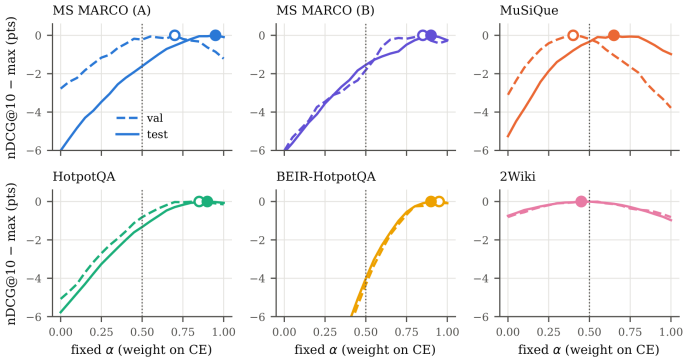


Figure 1: nDCG@10 as a function of the fixed weight α (min–max), relative to each curve's maximum; dashed = validation, solid = test, circles = arg-max. Left of the dotted line the contextual score has more weight. The single-hop reference (MS MARCO) peaks far to the right; MuSiQue (validation) and 2Wiki peak left of 0.5. The y-axis is clipped at -6 points.

- Each curve: nDCG@10 of the fixed fusion as α goes from 0 (GAT only) to 1 (CE only).
- Single-hop **MS MARCO** peaks far right: $\alpha = 0.70$–0.95.
- Multi-hop **MuSiQue** (val, $\alpha = 0.40$) and **2Wiki** ($\alpha = 0.45$) peak **left of 0.5**: the contextual score gets more weight.

1. More hops $\Rightarrow$ lower α (MuSiQue)

		nDCG@10 of each signal alone				Best fixed α						% prefer ctx	
		val		test		val			test				
Hops	n (test)	CE	GAT	CE	GAT	mm	z	rank	mm	z	rank	val	test
2-hop	1,252	76.9	76.8	73.8	68.0	**0.40**	0.50	**0.45**	0.80	0.80	0.95	48	40
3-hop	760	69.2	71.0	71.8	66.7	**0.40**	**0.40**	**0.30**	0.55	0.65	0.85	54	39
4-hop	405	66.1	71.7	61.3	63.3	**0.25**	**0.30**	**0.00**	**0.40**	**0.40**	**0.00**	60	52

Table 3: MuSiQue by hop count. Left: the cross-encoder loses 10.8 (val) and 12.5 (test) nDCG@10 points from 2 to 4 hops, the GAT only 5.1 and 4.7, and overtakes it on 4-hop questions. Middle: the best fixed α falls with the hop count under every scale-free normaliser (mm = min–max, z = z-score); bold = below 0.5, i.e. the contextual score dominates. Right: share of α-sensitive questions whose optimal α-interval is centred below 0.5. Tie-robust per-question test of the trend: Spearman(preferred α, hops) = -0.072 [-0.110, -0.032] on val ($p = 0.0005$) and -0.068 [-0.110, -0.025] on test ($p = 0.004$).

Best α falls with each hop. Test, min–max: **0.80** $\rightarrow$ **0.55** $\rightarrow$ **0.40**; val: 0.40 $\rightarrow$ 0.40 $\rightarrow$ 0.25.

Why: 2 $\rightarrow$ 4 hops, the cross-encoder drops 12.5 points (test), the GAT only 4.7.

1. The trend holds under every normaliser

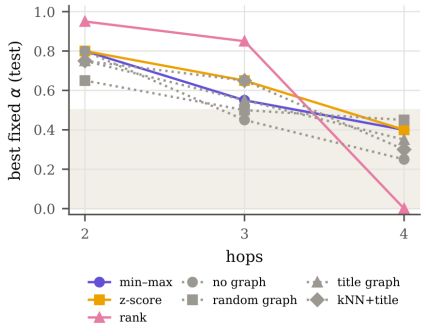


Figure 3: MuSiQue: best fixed α on test per hop count under the three scale-free normalisers (coloured) and for GATs retrained on four other graphs (grey, dotted; min–max). Shaded: $\alpha < 0.5$. Every setting declines from 2 to 4 hops, and every 4-hop optimum is below 0.5.

- Min–max, z-score and rank all give a **monotone decline** of the best α from 2 to 4 hops.
- 4-hop questions sit **below 0.5** in all 14 normaliser / graph / split settings.
- Per question: preferred α correlates negatively with hop count (Spearman $\rho \approx -0.07$, $p \leq 0.004$ on val and test).

1. Same pattern by question type (2Wiki)

	nDCG@10 (test)		Best fixed α		% prefer	Learned Q gate: mean α and ΔnDCG@10 vs. fixed (test)					
Type	CE	GAT	val	test	ctx	min–max		z-score		rank	
comparison	97.3	95.5	0.90	0.65	30	0.72	+0.10	0.52	-0.04	0.59	-0.07
compositional	88.5	88.9	**0.45**	0.50	53	0.45	-0.01	0.34	-0.08	0.44	+0.25$^{\blacktriangle}$
inference	86.7	87.6	**0.25**	**0.45**	59	0.30	-0.23$^{\triangledown}$	0.22	-0.03	0.38	+0.27
bridge-comparison	90.1	92.0	**0.25**	**0.20**	67	0.35	+0.16$^{\blacktriangle}$	0.33	+0.22$^{\blacktriangle}$	0.17	+0.81$^{\blacktriangle}$

Table 4: 2WikiMultiHopQA by question type, ordered from the simplest (comparison: both entities named in the question) to the most compositional (bridge-comparison). Bold $\alpha < 0.5$. The query-only gate, which never sees the type label, gives comparison questions the highest α under all three normalisers, and gains significantly on bridge-comparison questions ($^{\blacktriangle}$/$^{\triangledown}$: paired bootstrap $p < 0.05$).

Comparison:
$\alpha = 0.65$–0.90,
cross-encoder wins.

Inference, bridge-comp.:
$\alpha = 0.20$–0.45,
GAT wins.

Q gate, blind to the type:
highest α on comparison,
gains on bridge-comp.

2. Adaptive α gains on complex queries

Norm.	Gate	MSM-A	MSM-B	MuSiQue	HotpotQA	BEIR-HQA	2Wiki
none	Q	-0.48$^{\triangledown}$	+0.01	+0.00	-0.00	-0.00	+0.00
	QS	-0.48$^{\triangledown}$	+0.00	+0.00	+0.01	-0.00$^{\triangledown}$	+0.00
	S	-0.49$^{\triangledown}$	+0.00	+0.02	+0.01	-0.00	-0.00
minmax	Q	**+0.00**$^{\blacktriangle}$	+0.00	+0.00	-0.02	-0.00	+0.03
	QS	+0.00	+0.01	-0.01	+0.02	+0.00	-0.12$^{\triangledown}$
	S	+0.00	**+0.02**$^{\blacktriangle}$	-0.00	+0.04	+0.01	+0.00
	QS+C	–	–	+0.00	+0.03	-0.00	-0.11$^{\triangledown}$
zscore	Q	+0.01	**+0.05**$^{\blacktriangle}$	-0.15$^{\triangledown}$	+0.06	**+0.18**$^{\blacktriangle}$	+0.00
	QS	**+0.02**$^{\blacktriangle}$	-0.01	-0.14$^{\triangledown}$	+0.00	-0.01$^{\triangledown}$	-0.00
	S	+0.01	+0.01	-0.12$^{\triangledown}$	+0.01	-0.01	-0.01$^{\triangledown}$
rank	Q	+0.01	+0.02	**+0.43**$^{\blacktriangle}$	-0.01	-0.01	**+0.30**$^{\blacktriangle}$
	QS	-0.12	-0.04	**+1.22**$^{\blacktriangle}$	-0.04	-0.04$^{\triangledown}$	**+0.41**$^{\blacktriangle}$
	S	-0.10	-0.02	**+1.37**$^{\blacktriangle}$	-0.00	-0.02$^{\triangledown}$	**+0.19**$^{\blacktriangle}$

Table 6: Every adaptive gate against the best fixed α under the same normaliser: test ΔnDCG@10 $\times 100$ (paired bootstrap; $^{\blacktriangle}$ significantly better, $^{\triangledown}$ worse, $p < 0.05$ uncorrected).

7 of the 11 significant gains are on **multi-hop** sets; largest on MuSiQue (rank): +0.43 / +1.22 / +1.37.

MS MARCO gains ≤ 0.05. Oracle per-query α headroom: 7.4 (MuSiQue) vs. 5.0 (MS MARCO).

2. Transfer: α follows the target data

			Transferred gate, test nDCG@10				In-domain		Q gate
Setting	Train	Test	fixed	Q	QS	S	fixed	α_0	$\bar{\alpha}$ (sd)
leave-one-out	MSM+HQA	MuSiQue	71.99	71.93	71.98	72.00	71.28	0.70	0.73 (0.02)
mixed	MSM+MuSiQue+HQA	MS MARCO (A)	40.13	40.29	40.18	40.22	40.89	0.55	0.59 (0.12)
mixed	MSM+MuSiQue+HQA	HotpotQA	88.88	89.44	89.32	89.34	89.89	0.55	0.68 (0.15)
mixed	MSM+MuSiQue+HQA	MuSiQue	71.98	71.77	71.81	71.79	71.28	0.55	0.54 (0.16)
zero-shot	MSM	BEIR-HotpotQA	71.45	71.45	71.45	71.46	72.63	0.70	0.70 (0.00)
zero-shot	MSM	HotpotQA	89.24	89.24	89.25	89.20	89.89	0.70	0.70 (0.00)
zero-shot	MSM	2Wiki	90.99	90.99	90.99	90.99	91.75	0.70	0.70 (0.00)

Table 9: Transfer and mixed training of the gate (E4, min–max, $\times 100$, 5 seeds). Zero-shot uses the frozen MS MARCO GAT for s_{ctx}; mixed and leave-one-out use each dataset's own GAT. α_0 is the fixed α the gate starts from (selected on the training sources' val).

One gate on mixed data gives MuSiQue the **lowest** mean α (0.54) vs. MS MARCO 0.59 and HotpotQA 0.68.

Zero-shot MS MARCO gates keep $\alpha = 0.70$ and lose to the target's own setting, e.g. 2Wiki 90.99 vs. 91.75.

Conclusion

- **Alpha follows the hops**: the more reasoning steps a query needs, the more weight belongs on the graph-contextual score.
- MuSiQue: best α 0.80 $\rightarrow$ 0.55 $\rightarrow$ 0.40 for 2 $\rightarrow$ 3 $\rightarrow$ 4 hops; 4-hop below 0.5 in every setting.
- Mechanism: the cross-encoder degrades with hops much faster than the GAT.
- One fixed weight is a compromise; query-dependent gates gain mostly on multi-hop data.

Limits

HotpotQA keeps a high α (single-hop shortcuts). Min–max gates mostly stay at their start value.

Next

Gates with explicit complexity signals (predicted hop count), and fusing at the passage-pair level.

Thank you

Questions?

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